

Autodesk® Scaleform®

HUD

Scaleform 4.2 HUD

Flash UI

C++

AAA

Nate Mitchell, Prasad Silva

2.0

2010 7 30

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Autodesk® Scaleform® 4.2

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Autodesk Scaleform

HUD

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

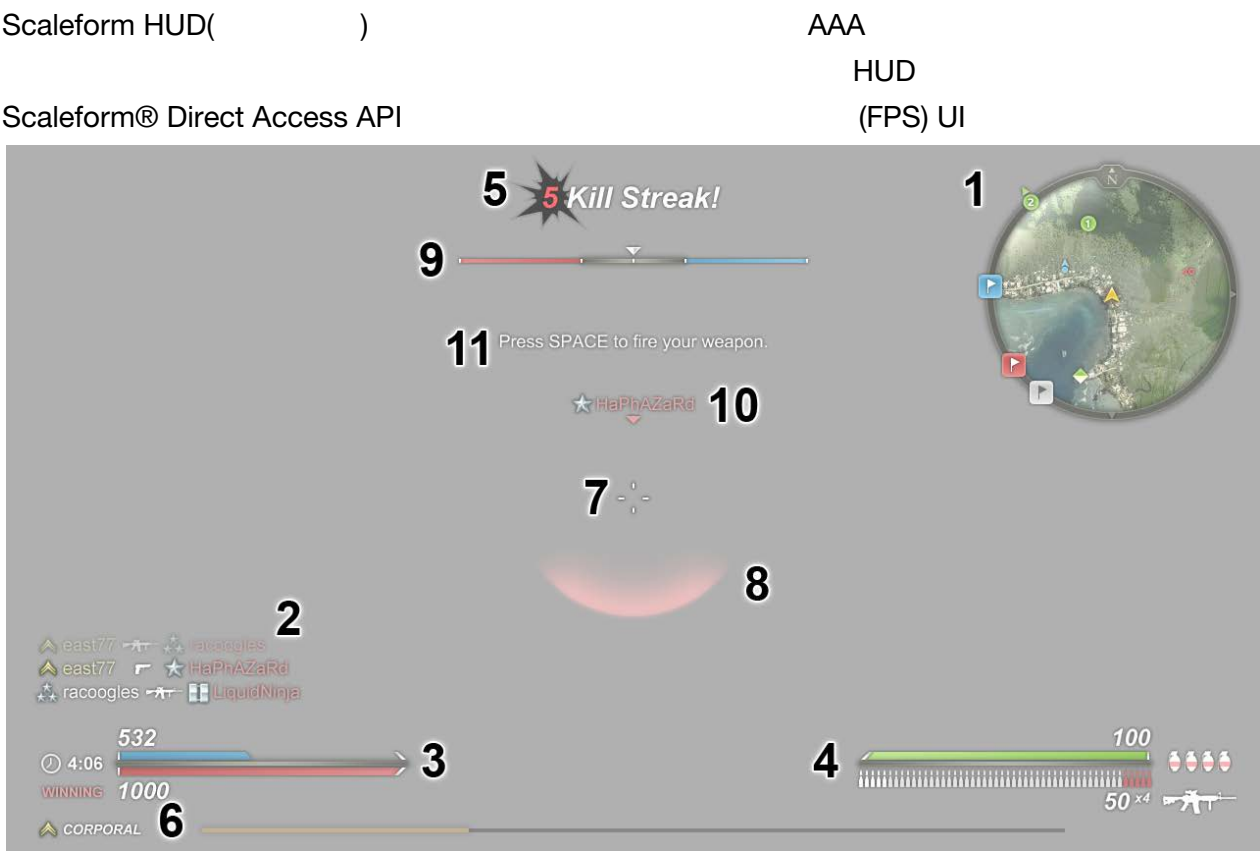
(301) 446-3200

(301) 446-3199

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1 引言



1 HUD

- HUD
- UI :
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 - 9.
 10. /
 - 11.

FPS

2 概述

2.1 檔位置和生成 (Build) 注釋

:

- *Apps\Kits\HUD\ - HUD C++*
- *Bin\Data\AS2 or AS3\Kits\HUD\ - Flash ActionScript (AS2 ActionScript 2 Flash 8 , AS3 ActionScript 3 Flash 10) 。*
- *Projects\Win32\{Msvc80 or Msvc90}\Kits\HUD - Windows Visual Studio 2005/2008*
- *Projects\Xbox360\{Msvc80 or Msvc90}\Kits\HUD - Xbox 360 Visual Studio 2005/2008*
- *Projects\Common\HudKit.mk - HUD PS3*

Bin\Kits\HUD (Windows) HUDKit.exe
Scaleform SDK

Windows , Projects\Win32\Msvc80\Kits(Msvc90\Kits) Scaleform 4.2 Kits.sln
Debugging() “Working ”

Bin\Data\AS2\Kits\HUD or Bin\Data\AS3\Kits\HUD

Xbox 360 , Projects\Xbox360\{Msvc80 Msvc90}\Kits\HUD Scaleform 4.2 Kits.sln
(Bin\Data\AS2 or
As3\Kits\HUD) Xbox 360

PS3, Scaleform make , HUD
(HUD Kit) PS3 , SN Systems .

Bin\PS3 :

app_home/ Directory: {Local Scaleform Directory}\Bin\Data\AS2 or AS3\Kits\HUD

2.2 演示使用

HUD, :

() 3

1 1 500 , (15)

4 ,

(Power-up)

2.3 控制方案

2.3.1 鍵盤 (Windows)

W, A, S, D	WS A, D
SPACE	
R	
G	,
ESC	(Text Field) ,
B	AI
1, 2	1 2

2.3.2 (Xbox/PS3)

	W, S	A, D
X /		
		,
B /		,
Y /		

3 構

HUD Flash UI HUD C++ C++ ,
HUD

3.1 C++

C++ HUD MVC

, , Flash UI Fx ε Z 1

Flash UI Fx ε Z 1

3.1.3 戲

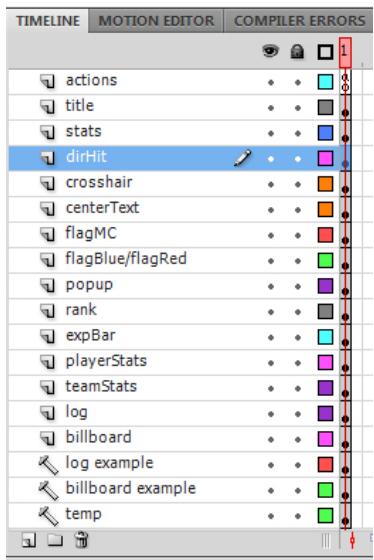
- HUDAdapter.h/.cpp – , HUD ,
- HUDEntityController.h/.cpp – , HUD ,
- HUDSimulation.h/.cpp – UI

3.2 Flash

HUD Flash :HUDKit.fla Minimap.fla FLA HUD
UI , Scaleform UI
 ,
 ,
Scaleform , C++ ActionScript Flash Flash , HUD
Classic Tweens Flash
GfX::Value::GotoAndPlay() C++

3.2.1 HUDKit.fla

Bin\Data\AS2 or AS3\Kits\HUD HUDKit.fla SWF HUDKitDemo
Flash UI , HUDKit.fla



2 HUDKit.fla

HUDKitDemo	Frame 1(1)	External Interface	HUDKit MovieClip:
ExternalInterface.call("registerHUDView", this);			

MovieClip()	,C++	MovieClip	C++ HUDView
--------------	------	-----------	-------------

“HUD ”	MovieClips	Scaleform C++
--------	------------	---------------

3.2.2 Minimap.fla

3 Minimap

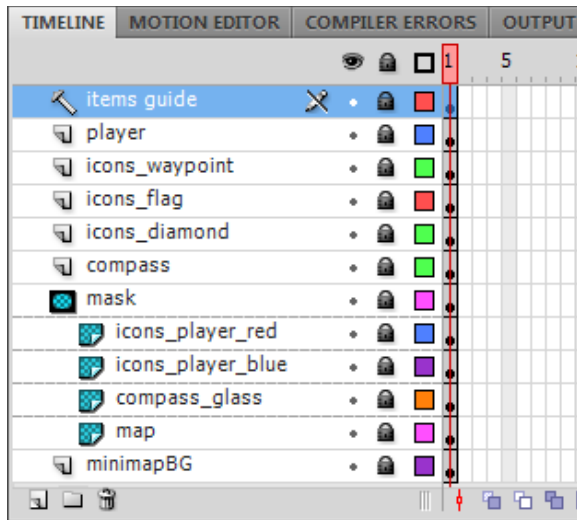
Minimap.fla ‘Minimap’ , , ,
 3D
 , (“ ”) ,
 :

- ()
- ()
- ((Flag) : , ,)
- ()

Direct Access API
 C++

MovieClips (Stage) MovieClips

‘Minimap’



4 'minimap'

- **items guide** / (Guide Layer) SWF
- **player**
- **icons_waypoint**
- **icons_flag**
- **icons_diamond**
- **compass** (“ ”)
- **mask** (Mask Layer)

- o **icons_player_red**
- o **icons_player_blue**
- o **compass_glass**
- o **map**

- **minimapBG**

‘Minimap’ com.scaleform.Minimap (Bin\Data\AS2 or AS3\Kits\HUD\com\scaleform
))
HUDKitDemo

4 HUD 視圖

```

C++ HUD          HUDKit.swf          :

• FxHUDView – HUD( HUDKit.swf) (Core Manager)
  MovieClips
• BillboardCache – / ( ,
  )
• FxHUDLog – HUD
  FxHUDMessages
• FxHUDMessage – ,

, FxHUDView MC(MovieClip )

, FxHUDView UpdateView() MovieClip

HUDKit fla UI

FxHUDView::UpdateView()
:UpdateTeamStats() UpdatePlayerEXP() Update PlayerStats() UpdateEvents() UpdateTutorial()
UpdateBillboards() ,
UI

, UpdateView() ,

Scaleform Direct Access API, Flash
, Gfx::Value Scaleform Flash
Gfx::Value::SetDisplayInfo() Gfx::Value::GetDisplayInfo() Gfx::Value::SetText() HUD
,

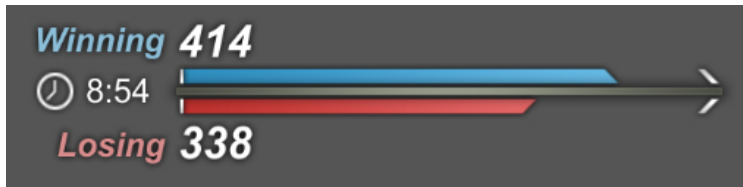
```

4.1 回合統計資料和記分牌

```

teamStats MovieClip( HUDKit fla 1 (Scene 1) teamStats )
(redWinning, blueWinning) (scoreRed, scoreBlue)
(teamRed, teamBlue), (roundTime)

```



5 teamStats

FxHUDView::UpdateTeamStats()

teamStats

teamStats MovieClip

```
void FxHUDView::UpdateTeamStats(FxHUDEnvironment *penv)
{
    // Retrieve the scores from the simulation.
    unsigned scoreBlue = unsigned(penv->GetHUDBlueTeamScore());
    unsigned scoreRed = unsigned(penv->GetHUDRedTeamScore());

    String text;
    Value tf;
    Value::DisplayInfo info;

    // We won't update any element who has not changed since the last update
    // to avoid unnecessary updates. To do so, we compare the previous
    // simulation State to the latest simulation data. If the two are different,
    // update the UI element in the HUD View.
    if (State.ScoreRed != scoreRed)
    {
        Format(text, "{0}", scoreRed);
        ScoreRedMC.SetText(text); Update the red team score text.

        info.SetScale((scoreRed / (Double)MaxScore) * 100, 100);
        TeamRedMC.SetDisplayInfo(info); // Update and scale the red team score
                                        bar movieClip.
    }

    if (State.ScoreBlue != scoreBlue)
    {
        Format(text, "{0}", scoreBlue);
        ScoreBlueMC.SetText(text); // Update the blue team score text.

        info.SetScale((scoreBlue / (Double)MaxScore) * 100, 100);
        TeamBlueMC.SetDisplayInfo(info); // Update and scale the blue team
                                        score bar movieClip.
    }

    if (State.ScoreRed != scoreRed || State.ScoreBlue != scoreBlue)
    {
        // Update the "Winning" and "Losing" text fields for both teams
        // if the score has changed.
    }
}
```

```

        if (scoreBlue > scoreRed)
        {
            RedWinningMC.SetText(LosingText);
            BlueWinningMC.SetText(WinningText);
        }
        else
        {
            RedWinningMC.SetText(WinningText);
            BlueWinningMC.SetText(LosingText);
        }
    }

    // Update the roundtime clock
    float secondsLeft = penv->GetSecondsLeftInRound();
    unsigned sec = ((unsigned)secondsLeft % 60);
    if (sec != State.Sec)
    {
        unsigned min = ((unsigned)secondsLeft / 60);
        String timeLeft;
        Format(timeLeft, "{0}:{1:0.2}", min, sec);
        RoundTimeMC.SetText(timeLeft);
        State.Sec = sec;
    }
}

```

```

        Scaleform::String , Gfx::Value::SetText() scoreRed scoreBlue
teamRed teamBlue X Gfx::Value::SetDisplayInfo()
0% ,

blueWinning/redWinning ,
Gfx::Value::SetText() HUDView "Winning" "Losing"
textField , UpdateView()
, Gfx::Value::SetText()

roundTime textField

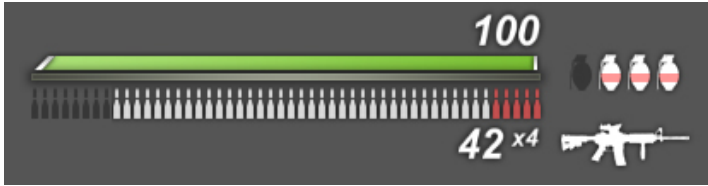
```

4.2 Player Stats Ammo

```

playerStats ( HUDKit.flas 1 playerStats ) (healthN,
health) (weapon) HUD (reload)

```

6 playerStats

, clipN ammoN ammo ammoBG grenade grenadeBG

clipN ammoN

ammo/healthVehicle :ammo ammoBG,

51 Keyframes(),

GfX::Value::GotoAndStop()

,

- *ammo*
- *ammoBG*

grenade/armor ammo , :grenade grenadeBG,

5

1 4 , 5 [GfX::Value::GotoAndStop\(\)](#)

,

- *grenade*
- *grenadeBG*

playerStats

`void FxHUDView::UpdatePlayerStats(FxHUDEnvironment *penv)`

{

`FxHUDPlayer* pplayer = penv->GetHUDPlayer();`

`// Load latest player info.`

`unsigned ammoInClip = pplayer->GetNumAmmoInClip();`

`unsigned ammoClips = pplayer->GetNumClips();`

`unsigned clipSize = pplayer->GetMaxAmmoInClip();`

`unsigned grenades = pplayer->GetNumGrenades();`

`unsigned ammoFrames = 51;`

`unsigned grenadeFrames = 5;`

`String text;`

```

if(ppplayer->GetHealth() != State.Health)
{
    unsigned health = UInt(ppplayer->GetHealth() * 100);
    Format(text, "{0}", health);
    HealthNMC.SetText(text); // Update the health text field.

    Value::DisplayInfo info;
    info.SetScale(Double(health), 100);
    HealthMC.SetDisplayInfo(info); // Update and scale the health bar.
}

if(ammoInClip != State.AmmoInClip)
{
    Format(text, "{0}", ammoInClip);
    AmmoNMC.SetText(text); // Update the ammo text field.

    // The ammo is divided into two parts: the white bullet icons (AmmoMC)
    // and their respective backgrounds (AmmoBGMC).

    // To update the number of bullets currently in the clip, we use
    // GotoAndStop with AmmoMC and select the frame with the proper number
    // of white bullet icons. There are 51 frames in AmmoMC (defined above
    // in ammoFrames) and the first frame displays every bullet icon.

    // To display X number of bullets, we subtract X from the total number
    // of frames in AmmoMC (51) and GotoAndStop on the result.
    AmmoMC.GotoAndStop(ammoFrames - ammoInClip);
}

if(ammoClips != State.AmmoClips)
{
    Format(text, "x{0}", ammoClips);
    ClipNMC.SetText(text); // Update the remaining ammo clips text field.
}

if(grenades != State.Grenades)
{
    // Grenades are setup similar to Ammo. The first frame for the
    // Grenades movieClip shows 4 white grenade icons. The second frame
    // shows 3.

    // To display X number of grenades, we subtract X from the total number
    // of frames in GrenadeMC (5) and GotoAndStop on the result.
    // Note: GrenadeMC actually contains 10 frames but 6-10 are unused by
    // this implementation.
    GrenadeMC.GotoAndStop(grenadeFrames - grenades);
}

```

```

if (pplayer->GetWeaponType() != State.weaponType)
{
    Change the weapon icon if the player has changed weapons.
    unsigned weaponFrame = GetWeaponFrame(pplayer->GetWeaponType());
    WeaponMC.GotoAndStop(weaponFrame);

    // Update the ammo background icons based on clipSize.
    if (clipSize != State.AmmoClipSize)
        AmmoBGMC.GotoAndStop(ammoFrames - clipSize);
}

// If the ammo in clip is less than 6 bullets, play the "Reload" indicator
// movieClip.
if (ammoInClip <= 5)
{
    // ReloadMC will play until we GotoAndStop on Frame 1, so no need to
    // start the animation more than once.
    if (!bReloadMCVisible)
    {
        bReloadMCVisible = true;
        ReloadMC.GotoAndPlay("on"); // Will cycle back to "on" frame when
                                    it reaches the end and play again.
    }
}

// If the reload indicator is displayed but there are more than six bullets
// hide it and stop the animation by playing frame 1 of the movieClip.
else if (bReloadMCVisible)
{
    ReloadMC.GotoAndStop("1"); // Frame 1 contains stop();
    bReloadMCVisible = false;
}
}

ammo , 1 50 , 51 0 ,
ammoMC.GotoAndStop(ammoFrames - ammoinClip) , HUD

ammoBG, ammoBGMC.GotoAndStop() X ,X
, (Machine Gun) 50
,ammoBG.GotoAndStop(1) 1 50
FxHUDView::UpdateView() MovieClip

```

4.3 旗標捕獲指示器

```
flagMC ( HUDKit fla 1 ) flagMC flag ,
: (arrow), CaptureState( )
, , ,
```



7 flagMC

```
flagMC Classic Tween Flash flagMC
4 , 4 :Hidden( ) Visible( ) Fade-In( ) FadeOut(
) GotoAndStop() Keyframe / (1 5 "On" "Off" )
FxHUDView::UpdateEvents() flagMC arrow( )
```

```
// Flag capture indicators and reticule behavior.
void FxHUDView::UpdateEvents(FxHUDEnvironment *penv)
{
    Value::DisplayInfo info;

    if (penv->IsHUDPlayerCapturingObjective())
    {
        if(!bFlagMCVisible)
        {
            SetVisible(&FlagMC, true); // Set the Flag MovieClip's visibility
                                     // to true
            FlagMC.GotoAndPlay("on"); // Play the fade-in animation once.
            bFlagMCVisible = true;
        }

        // Shift the x-coordinate of the Flag Arrow to show the capture state
        // of the flag the player is currently capturing.
        info.SetX(penv->GetHUDCaptureObjectiveState() * 177);
        FlagArrowMC.SetDisplayInfo(info);
        info.Clear();
    }
    else if (bFlagMCVisible & !penv->IsHUDPlayerCapturingObjective())
    {
        // If the flag indicator is still visible and the player is
        // no longer capturing an objective, play the fade-out animation
        FlagMC.GotoAndPlay("off");
    }
}
```

```

        bFlagMCVisible = false;
    }
}

flagMC.GotoAndPlay("on") ,flagMC , Visible( )
flagMC.GotoAndPlay("off") ,flagMC 11-20, 20 ,
1 Flash ; ActionScript C++ 1 stop();
, , 1 ,
Alpha 0

arrow Gfx::Value::setDisplayInfo\(\)
Gfx::Value::DisplayInfo, CaptureState X

```

4.4 武器十字線

```

Reticule( ) ( HUDKit.fla 1 ) reticule
:left right bottom top,
PlayerFire

```



8 reticule

```

PlayerFire :FxHUDView::UpdateEvents() FxHUDView::OnEvent()
,ReticuleHeat ReticuleFiringOffset FxHUDEvent

// If the player is firing his/her weapon, update the reticule's heat.
// The reticule elements are updated in FxHUDView::UpdateEvents() based on
// the ReticuleHeat and the ReticuleFiringOffset.
case FxHUDEvent::EVT_PlayerFire:
{
    if (ReticuleHeat < 8)
        ReticuleHeat += 2.5f;

    ReticuleFiringOffset = 2;
}

```

```

FxHUDView::UpdateEvents()
    , SetDisplayInfo()
0, 0 X Y , SetDisplayInfo()
GfX::Value::DisplayInfo
GfX::Value::SetDisplayInfo() DisplayInfo.Clear() DisplayInfo

```

```

// Shift the XY-coordinates of the reticule elements based on the
// firing duration and rate of fire.

```

```

if (ReticuleHeat > 0 || ReticuleFiringOffset > 0){
    if (ReticuleHeat > 1.0f)
        ReticuleHeat -= .02f;

    ReticuleFiringOffset -= .02f;
    if (ReticuleFiringOffset < 0)
        ReticuleFiringOffset = 0;

    info.SetY((-6) - (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Top.SetDisplayInfo(info);
    info.Clear();

    info.SetX((6) + (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Right.SetDisplayInfo(info);
    info.Clear();

    info.SetX((-6) - (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Left.SetDisplayInfo(info);
    info.Clear();

    info.SetY((6) + (ReticuleFiringOffset + ReticuleHeat));
    ReticuleMC_Bottom.SetDisplayInfo(info);
    info.Clear();
}

```

```

, , , ,

```

4.5 定向命中指示器

```

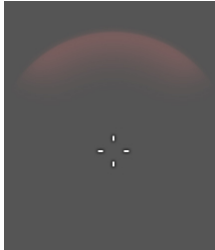
dirHit ( HUDKit.flas 1 dirHit ) dirHit
:tl l bl b br r tr t

```

Alpha , :

a)

b)



9 dirHit

Classic Tween Alpha , ~1 Flash
Gfx::Value::GotoAndPlay("on"),

```
void FxHUDView::OnEvent( FxHUDEvent* pevent )
{
    switch ( pevent->GetType() )
    {
        // If the Damage Event (Player was hit) is fired, show the appropriate
        // directional hit indicator.
        // We can use GotoAndPlay("on") to play the fade-in animation. It will
        // fade out on its own after ~1 second (this animation is setup on the
        // Flash timeline)
        case FxHUDEvent::EVT_Damage:
        {
            FxHUDDamageEvent* peventDamage = (FxHUDDamageEvent*)pevent;
            float dir = peventDamage->GetDirection();
            if (dir > 0)
            {
                if (dir > 90)
                {
                    if (dir > 135)
                        DirMC_B.GotoAndPlay("on");
                    else
                        DirMC_BR.GotoAndPlay("on");
                }
                else
                {
                    if (dir > 45)
                        DirMC_TR.GotoAndPlay("on");
                    else
                        DirMC_T.GotoAndPlay("on");
                }
            }
        }
    }
}
```

```

    }
    else
    {
        if (dir < -90)
        {
            if (dir < -135)
                DirMC_B.GotoAndPlay("on");
            else
                DirMC_BL.GotoAndPlay("on");
        }
        else
        {
            if (dir < -45)
                DirMC_L.GotoAndPlay("on");
            else
                DirMC_TL.GotoAndPlay("on");
        }
    }
    break;
}
}
}

```


4.6 等級和經驗顯示

```
expBar ( HUDKit.fla Scene 1 expBar ) ,
        expBar exp MovieClip,

rank ( HUDKit.fla Scene 1 rank ) rank 18 Keyframe,
      Keyframe
```



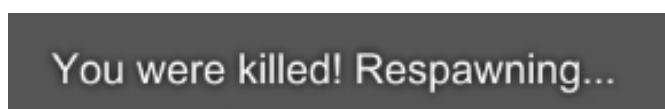
10 expBar rank

```
void FxHUDView::UpdatePlayerEXP(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
    if (pplayer->GetLevelXP() != State.Exp)
    {
        Value::DisplayInfo info;
        // Update the rank icon. Each rank icon is on a different frame of
        // the rank movieClip.
        RankMC.GotoAndStop(UInt(pplayer->GetRank() + 1));
        info.SetScale(pplayer->GetLevelXP() * 100, 100); // 伸縮 EXP 進度條
                                                         // 電影剪輯。
        ExpBarMC.SetDisplayInfo(info);
    }
}
```

```
,
,exp rank,
GFx::Value::GotoAndStop() X, X + 1( 0)
```

4.7 文本通知

```
centerTextMC ( HUDKit.fla 1 centerText ) ,
               ,centerText ,
```



5 centerTextMC

```

centerTextMC      centerText      ,      textField
      centerTextMC      "on5"(
      5      )      "on"(      3      )
1      ,      ,centerTextMC      Alpha      0,      "on"      "on5"
TextFieldMC.SetText(RespawnText); // Set the text field of the Center Text.
CenterTextMC.GotoAndPlay("on"); // Play the fade-in animation, it will fade on its
                                own after ~3 sec.

      ,      centerTextMC      textField      ,      GotoAndPlay("on")
      Classic Tween      ~3

```

4.8 彈出消息和通知

```

popup      (      HUDKit fla      1      popup(      )      )      ,
      3      (      )      ,      popup      ,      3-5
C++

```



6 popup

```

Popup      Classic Tweens      popup
      popup      4 :actions popupNumber popupText popupBG popupBG
"      "      popupText      popupNumber      textField

      "on"      ,      GotoAndPlay()      ,
      ,      1      ,      Alpha      0

```

```

// If a KillStreak event is fired, display the kill streak pop-up
case FxHUDEvent::EVT_KillStreak:
{
    FxHUDKillStreakEvent* peventKS = (FxHUDKillStreakEvent*)pevent;

    // Format the number of kills
    String text;
    Format(text, "{0}", peventKS->GetSrcKillStreak());
    PTextFieldMC.SetText(text); // 设置 PopUp 的文本框。

```

```

        // Play the fade-in animation, it will fade on its own after ~4 sec.
        PopupMC.GotoAndPlay("on");
    }
    break;

    ,FxHUDEvent      KillStreak
    C++              popupText's textField      PTextFieldMC      ,      GotoAndPlay("on")
    popup

```

4.9 消息和事件日誌

```

log      (      HUDKit fla      Scene 1      log(      )      )      HUD
,
,
,
~3

```



13 log logMessage

```

        logMessage
        logMessage
    )

    FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip(),
    LogMessage MovieClip      Log
,

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()
{
    if (MessageMCs.IsEmpty())
    {
        // Request a new line in the SWF
        Value logline;
        LogMC.AttachMovie(&logline, "LogMessage", "logMessage"+NumMessages);
        FxHUDMessage::Movieclip* pmc = new FxHUDMessage::Movieclip(logline);
        MessageMCs.PushBack(pmc); // Add new Message MC to list of unused
                                   message movieClips.
    }
}

```



```

        info.SetY(yoff);
        data->GetMovieclip()->GetRef().SetDisplayInfo(info);
    }
    data = next;
    yoff += rowHeight;
}

// Layout management and animation
Value::DisplayInfo info;
info.SetY(LogOriginalY - (NumMessages * rowHeight));
LogMC.SetDisplayInfo(info);

// Remove dead messages
data = Log.GetFirst();
while (!Log.IsNull(data))
{
    FxHUDMessage* next = Log.GetNext(data);
    if (!data->IsAlive())
    {
        Log.Remove(data);
        NumMessages--;

        FxHUDMessage::Movieclip* pmc = data->GetMovieclip();
        pmc->SetVisible(false);
        MessageMCs.PushBack(pmc);

        delete data;
    }
    data = next;
}
}

Log                FxHUDMessage                0
FxHUDMessage        ,                Log                ,                FxHUDMessage MovieClips
MessageMCs          ,                FxHUDMessage,        Gfx::Value::DisplayInfo.SetY()
Gfx::Value::SetDisplayInfo()    MessageMC

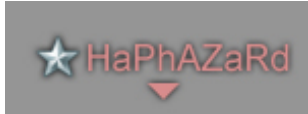
```

4.10 看板

```

billboard_container(    HUDKit fla    Scene 1    billboard    )
                        3D                                ,    3D
                        (Frustum)                        ,
                        ,

```



7 billboard_enemy

UI 2D ,

GfX::Value::SetDisplayInfo()

MovieClip (billboard_friendly, billboard_enemy)

billboard_label_(friendly/enemy) billboard_label_*
 textField billboard_friendly's "Show" Keyframe billboard_friendly
 Classic Tween "Show"(), ,

C++ FxHUDView::BillboardCache::GetUnusedBillboardMovieclip

:

```
if (UnusedBillboards.IsEmpty())
{
    Value::DisplayInfo info(false);

    Value billboard, temp;
    String instanceName;
    Format(instanceName, "{0}{1}", BillboardPrefix, BillboardCount++);
    CanvasMC.AttachMovie(&billboard, SymbolLinkage, instanceName);

    BillboardMC* pbb = new BillboardMC();
    pbb->Billboard = billboard;
    billboard.GetMember("label", &temp);
    temp.GetMember("textField", &temp);
    pbb->Textfield = temp;

    pbb->Billboard.SetDisplayInfo(info);
    UnusedBillboards.PushBack(pbb);
}
```

BillboardCache , CanvasMC
 , BillboardCache billboard_enemy/billboard_friendly
 CanvasMC.AttachMovie() , ,

```

Billboard                                :UpdateBillboards() BeginProcessing() EndProcessing()
GetUnusedBillboardMovieclip()           BeginProcessing()
EndProcessing()

```

```

UpdateBillboards()

```

```

// Process friendlies
// Friendly billboards will always show names

```

```

BillboardMC* pbb = NULL;

```

```

FriendlyBillboards.BeginProcessing();
for (unsigned i=0; i < friendlies.GetSize(); i++)
{
    info.Clear();
    if (FriendlyBillboards.GetUnusedBillboardMovieclip(friendlies[i].pPlayer,
                                                         &pbb))
    {
        info.SetVisible(true);
        pbb->Billboard.GotoAndPlay("show");
        // Load player info
        String title;
        Format(title,
               "<img src='rank{0}' width='20' height='20' align='baseline'
               vspace='-7' /> {1}",
               friendlies[i].pPlayer->GetRank()+1,
               friendlies[i].pPlayer->GetName());
        pbb->Textfield.SetTextHTML(title);
    }
    info.SetX(friendlies[i].X);
    info.SetY(friendlies[i].Y);
    info.SetScale(friendlies[i].Scale, friendlies[i].Scale);
    pbb->Billboard.SetDisplayInfo(info);
}

```

```

FriendlyBillboards.EndProcessing();

```

```

UpdateBillboards()
GotoAndPlay("show") , GFx::Value::SetDisplayInfo() MovieClip's X Y ,
GFx::Value::SetText()

```

```

textField      htmlText
src='rank{0}' HTML , + 1 (friendlies[i].pPlayer->GetRank()+1)

```

```
img                                     vspace {1}  
, (friendlies[i].pPlayer->GetName())
```


5 迷你地圖視圖

(Minimap) C++ (Apps\Kits\HUD\):

- **FxMinimap.h**
- **FxMinimapView.h/.cpp**

5.1 迷你地圖視圖

:

- **FxMinimapIconType** , ID
- **FxMinimapIcon**
- **FxMinimapIconCache** / /
 , MinimapIconFactoryBase
 , MovieClips
 Update() ,
 , : textField :
- o *PlayerIcon* ,
- o *FlagIcon*
- o *ObjectiveIcon*
- o *WaypointIcon*

- **FxMinimapView** UpdateView()
 PlayerIcon::Update() (FxMinimapView.cpp; 116) Direct

Access API MovieClip MovieClip
SetDisplayInfo (Alpha),
SetMember() ,GfX::Value::SetMember GfX::Movie::SetVariable :

```
virtual void Update(FxMinimapView* pview, FxMinimapEntity* pentity, float distSquared)
```

```
{
```

```
    SF_UNUSED(distSquared);
```

```
    bool hasIcon = (pentity->GetIconRef() != NULL);
```

```
    PointF pt = pentity->GetPhysicalPosition();
```

```
    bool showDir = pview->IsShowingPlayerDir();
```

```
    // If entity did not have an icon attached before, then wait for the
```

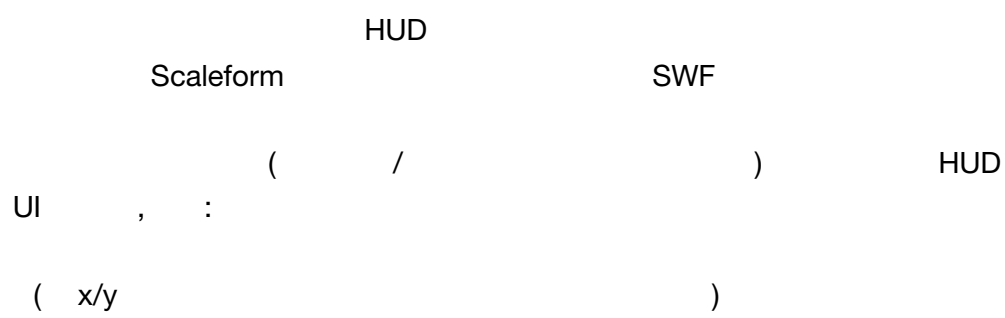
```

// next update to set the state data. picon is most probably not initialized
// (the movie needs to advance after creation)
if (hasIcon && (State != (int)showDir))
{
    // state 0: no arrow, state 1: show arrow
    Value frame(Double(showDir ? 2.0 : 1.0));
    MovieClip.Invoke("gotoAndStop", NULL, &frame, 1);
    State = (int)showDir;
}

pt = pview->GetIconTransform().Transform(pt);
Value::DisplayInfo info;
if (State)
{
    info.SetRotation(pview->IsCompassLocked() ?
        (pentity->GetDirection() + 90) : (pentity->GetDirection() -
            pview->GetPlayerDirection()));
}
info.SetPosition(pt.x, pt.y);
MovieClip.SetDisplayInfo(info);
}

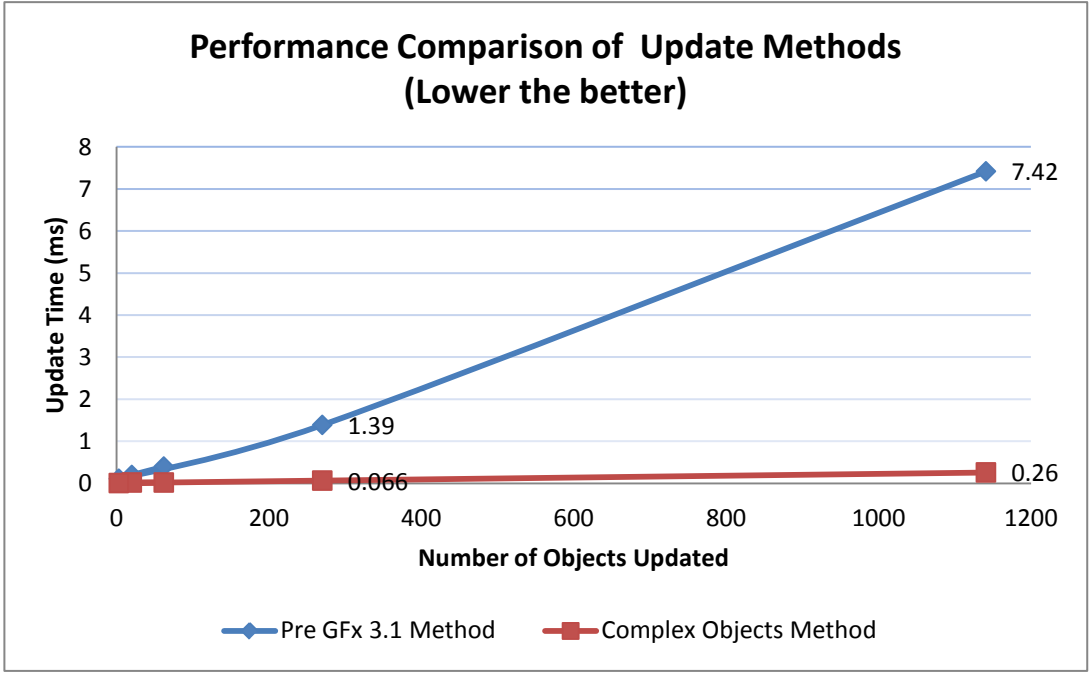
```

6 性能分析



6.1 性能統計資料

Scaleform, C++ GfX::Movie::SetVariable, GfX::Movie::Invoke, Movie, Direct Access API, HUD, Scaleform 3.1, Direct Access API, SetVariable/Invoke, 10-25 :



8 Direct Access API SetVariable/Invoke

,Direct Access
– HUD UI

1

Scaleform 15 HUD
Flash HUD :

	FPS				
Windows Vista MacBook Pro	1148	0.018	0.512	0.017	0.527
Xbox 360	1136	0.032	0.454	0.010	0.497
PlayStation 3	752	0.044	0.668	0.021	0.733

1 Scaleform HUD

1 ,HUD , 1 –
HUD UI

6.2 記憶體明細表

HUD

GFxExport

.GFX

1.

GFx

1858kb

- HUDKit.gfx - 33kb
- Minimap.gfx - 50kb
- fonts_en.gfx - 46kb
- gfxfontlib.gfx - 100kb

- HUDKit.gfx
 - o UI - 603kb
 - o - 72kb
 - o - 51kb
 - HUD , 43kb
- Minimap.gfx
 - o - 644kb
- Map.jpg - 259kb